QUALITY INFORMATION DOCUMENT

For OMI_CLIMATE_OSC_MEDSEA

Issue: 1.1

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	16/06/2023	All	Creation of the document	Pietro Miraglio Ali Aydogdu	Emanuela Clementi (Med-MFC Deputy)
1.1	28/05/2025	All	Extension of the time series and Setup of operational extension	Lucia Gualtieri	Emanuela Clementi

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1) INTRODUCTION

a) Product Name

OMI_CLIMATE_OSC_MEDSEA_volume_mean

b) Indicator Evaluation Method

Ocean salt content (OSC) is defined and represented here as the volume average of the integral of salinity in the Mediterranean Sea from z_1 = 0 m to z_2 = 300 m depth:

$$\overline{S} = \frac{1}{V}\int_V S \, \mathrm{dV} \qquad [1]$$

Time series of annual mean values area averaged ocean salt content is provided for the Mediterranean Sea (30°N, 46°N; 6°W, 36°E) and is evaluated where the bathymetry is deeper than 300 m. The total volume estimated is approximately 5.7e+5 km^3.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
MEDSEA_MULTIYEAR_PHY_006_004 (MEDREA24)	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-MED-QUID-006-004.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-MED-PUM-006-004.pdf
GLOBAL_REANALYSIS_PHY_001_031 (GREP)	QUID : https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-GLO-QUID-001-031.pdfPUM : https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-GLO-PUM-001-031.pdf
INSITU_GLO_TS_REP_OBSERVATIONS_013_001_B (CORA)	QUID : https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-INS-QUID-013-001b.pdf PUM : https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-INS-PUM-013-001-b.pdf
MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012 (ARMOR3D)	QUID : https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-MOB-QUID-015-012.pdf PUM : https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-MOB-PUM-015-012.pdf

2) SCIENTIFIC VALIDATION

a) Mediterranean Ocean Salt Content (0-300 m)

More details and full scientific evaluation can be found in the Copernicus Marine Ocean State Report issue 7 (OSR7) in the contribution by Aydogdu et al. (2023). Note that the estimates in Aydogdu et al. (2023) are provided monthly while here we evaluate the results per year.
Ocean salt content (OSC) is defined and represented here as the volume average of the volume integral of salinity in the Mediterranean Sea from z_1 = 0 m to z_2 = 300 m depth:

$$\overline{S} = \frac{1}{V}\int_V S\,\mathrm{dV} \qquad [1]$$

The time series of the annual volume mean ocean salt content is calculated in the Mediterranean Sea (30°N, 46°N; 6°W, 36°E) and is evaluated only where the bottom topography is deeper than 300m.

The quality evaluation of OMI_CLIMATE_OSC_MEDSEA_volume_mean is based on the "multi-product" approach. Six global products and a regional (Mediterranean Sea) product have been used to build an ensemble mean and its associated ensemble spread (Figure 1, dark red line and shaded area). For each dataset, all available data is used thus resulting in a different time extension. Consequently, the ensemble is calculated over the time period common to all products, extending at least up to three years prior to the current year [−3Y] (or closer to the present, depending on data availability), and it is extended operationally on an annual basis. For the MEDREA, both reanalysis and interim data are used, thus the index is calculated up to the year preceding the current one [-1Y].
Results of Fig. 1 indicate how each member is distributed around the ensemble mean of 38.57 and show a continuous increase over the period 1993-2021 at rate of $7.0*10^{-3}$ $\pm 3.1*10^{-4}$ psu yr^{-1} in the upper 300m. During the early 1990s in the entire Mediterranean Sea, there is a large spread in salinity with the observational based datasets showing a higher salinity, while the reanalysis products present relatively lower salinity. The maximum spread between the period 1993–2021 occurs in the 1990s with a value of 0.12 psu, and it decreases to as low as 0.02 psu by the end of the 2010s. The global reanalysis products in general have a larger trend compared to the observational products. The regional MEDREA24 stays in between the two different types of estimates (i.e. the dynamical reanalysis-based and the observation-based salt content volume mean) and close to the ensemble mean.

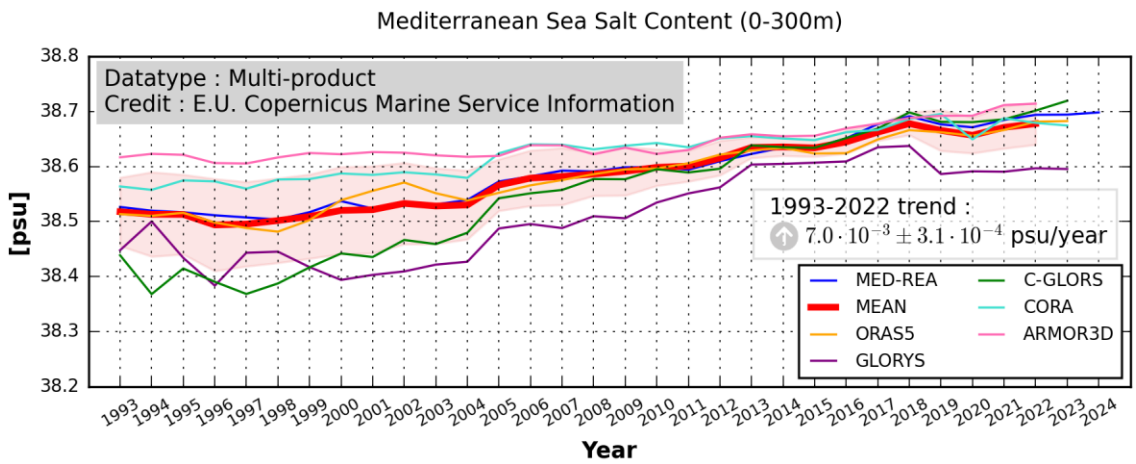

Figure 1: *Time series of volume averaged ocean salt content in the Mediterranean Sea integrated over the 0-300m depth layer. Time series are based on different data products, i.e., Mediterranean Sea Reanalysis MEDREA24 (dark blue), global ocean reanalysis GLORYS (purple), C-GLORS (green), ORAS5 (orange), FOAM (dark green); global observational based products CORA (light blue), ARMOR3D (pink), the ensemble mean (dark red) and the ensemble spread (red shaded). Details on the products are given in the corresponding PUM for this OMI.*

3) REFERENCES

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